

5. Derivatives of Inverse Trigonometric Function

Derivatives of Inverse Trigonometric Functions

$$\begin{aligned} \textcircled{1} \quad \frac{d}{dx}(\sin^{-1}x) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\csc^{-1}x) &= -\frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}(\cos^{-1}x) &= -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\sec^{-1}x) &= \frac{1}{x\sqrt{x^2-1}} \\ \textcircled{7} \quad \frac{d}{dx}(\tan^{-1}x) &= \frac{1}{1+x^2} & \frac{d}{dx}(\cot^{-1}x) &= -\frac{1}{1+x^2} \end{aligned}$$

use implicit differentiable
+
trigonometric identity.

Derive

①. $\frac{d}{dx}(\sin^{-1}x)$ $\frac{d}{dx}(\cos^{-1}x)$ is the same

Denote $y = \sin^{-1}x \Rightarrow x = \sin y. / ()'$

$$1 = \frac{dy}{dx} \cdot \cos y \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

$$\cos y = \cos(\sin^{-1}x) = \sqrt{1 - \sin^2(\sin^{-1}x)} = \sqrt{1-x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

②. $\frac{d}{dx}(\tan^{-1}x)$

Denote $y = \tan^{-1}x \Rightarrow x = \tan y. / ()'$

$$1 = \frac{dy}{dx} \cdot \sec^2 y$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1+x^2}$$

③. $\frac{d}{dx}(\csc^{-1}x)$

Denote $y = \csc^{-1}x \Rightarrow x = \csc y. / ()'$

$$1 = y'(\csc y)' = \frac{dy}{dx} \cdot -\csc y \cot y$$

$$\frac{dy}{dx} = -\frac{1}{\csc y \cot y} = -\frac{\sin y}{\cos y} = -\frac{\sin^{-1}y}{\sqrt{1-\sin^2 y}}$$

$$\sqrt{1-\sin^2 y} = \sqrt{1-\frac{1}{\csc^2 y}} = \sqrt{\frac{\csc^2 y - 1}{\csc^2 y}}$$

$$\frac{dy}{dx} = -\frac{1}{\csc y} \cdot \frac{\csc y}{\sqrt{\csc^2 y - 1}} = -\frac{1}{x\sqrt{x^2-1}}$$

④. $\frac{d}{dx}(\sec^{-1}x)$

$$y = \sec^{-1}x \Rightarrow x = \sec y. / ()'$$

$$1 = \frac{dy}{dx}(\sec y)' = \frac{dy}{dx} \cdot \frac{\sin y}{\cos y}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sin y \sec y} = \frac{1}{\sqrt{\sec^2 y - 1}} \cdot \frac{1}{\sec y} \\ &= \frac{1}{x\sqrt{x^2-1}} \end{aligned}$$

⑤. $\frac{d}{dx}(\cot^{-1}x)$

Denote $y = \cot^{-1}x \Rightarrow x = \cot y. / ()'$

$$1 = \frac{dy}{dx}(\cot y)' = \frac{dy}{dx} \cdot (-\csc^2 y)$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{\csc^2 y} = -\frac{1}{1 + \cot^2 y} \\ &= -\frac{1}{1+x^2} \end{aligned}$$

The composition of the inverse trigonometric.

三角函数关系

	$\sin \theta$	$\cos \theta$	$\tan \theta$
$\arcsin x$	$\sin(\arcsin x) = x$	$\cos(\arcsin x) = \sqrt{1-x^2}$	$\tan(\arcsin x) = \frac{x}{\sqrt{1-x^2}}$
$\arccos x$	$\sin(\arccos x) = \sqrt{1-x^2}$	$\cos(\arccos x) = x$	$\tan(\arccos x) = \frac{\sqrt{1-x^2}}{x}$
$\arctan x$	$\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}$	$\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$	$\tan(\arctan x) = x$
$\text{arccot} x$	$\sin(\text{arccot} x) = \frac{1}{\sqrt{1+x^2}}$	$\cos(\text{arccot} x) = \frac{x}{\sqrt{1+x^2}}$	$\tan(\text{arccot} x) = \frac{1}{x}$
$\text{arcsec} x$	$\sin(\text{arcsec} x) = \frac{\sqrt{x^2-1}}{x}$	$\cos(\text{arcsec} x) = \frac{1}{x}$	$\tan(\text{arcsec} x) = \sqrt{x^2-1}$
$\text{arccsc} x$	$\sin(\text{arccsc} x) = \frac{1}{x}$	$\cos(\text{arccsc} x) = \frac{\sqrt{x^2-1}}{x}$	$\tan(\text{arccsc} x) = \frac{1}{\sqrt{x^2-1}}$

remember tips:

①. $\sin \arcsin x = x$

↓

$$\cos \arcsin x = \sqrt{1 - \sin^2(\arcsin x)}$$

$$= \sqrt{1 - x^2}$$

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$$\tan \arcsin x = \frac{x}{\sqrt{1-x^2}}$$

②. $\sin(\arccos x) \Rightarrow \cos(\arccos x) = x$

$$y = \arccos x \Rightarrow x = \cos y \Rightarrow \sqrt{1-x^2} = \sin y \Rightarrow \sin(\arccos x) = \sqrt{1-x^2}$$

③. $\tan(\arcsin x)$ $y = \arcsin x \Rightarrow x = \sin y$

$$\tan(\arcsin x) = \frac{\sin y}{\cos y} = \frac{x}{\sqrt{1-x^2}}$$